

KUKKA BHARATH

Faridabad, Haryana

[bharathkukka](#)

[bharathkukka](#)

[Portfolio](#)

bharathkukka86@gmail.com

[+91-9505666566](tel:+91-9505666566)

EDUCATION

Manav Rachna University

June 2022 – June 2026

B. Tech in Robotics and AI Engineering - **CGPA - 8.78**

Delhi NCR, India

Sri Chaitanya Junior College of Arts and Science

June 2020 – May 2022

Board of Intermediate Education - MPC - **Percentage - 94.4%**

Khammam, Telangana

COURSEWORK

- Data Structures & Algorithms
- Operating Systems
- Artificial Intelligence
- Control Systems
- OOPS Concept
- Mechatronics

TECHNICAL SKILLS

Programming Languages: Java, Python, MATLAB, Embedded C, VHDL

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, ROS 2, Robotics Toolbox

Developer Tools: PyCharm, Eclipse, MATLAB, VS Code, Arduino IDE **Version Control:** Git, GitHub

Specializations: Deep Learning (CNNs), NLP, Kinematics & Dynamics of Manipulators

Sensors & Actuators: Cameras, Encoders, Servo & Stepper Motors **Communication Protocols:** UART

PROJECTS

Emotion Recognition System | Python, TensorFlow, CNN, Transfer Learning

Jun 2024 – Present

- Developed a CNN model for emotion classification in psychological therapy, achieving **76% accuracy** through hyperparameter tuning and data augmentation, while exploring transfer learning to improve generalization.

Home Security System | Computer Vision, OpenCV, Python, Deep Learning

Jan 2025 – Present

- Developing a computer vision-based security system for real-time threat detection (**intruders, weapons, falls**) with an automated, multi-tiered emergency alert system.

4-Axis Manipulator | Python, Robotics, Kinematics, Control Systems

Jul 2024 – Dec 2024

- Engineered a 4-axis pick-and-place manipulator, derived kinematic equations for motion control, developed a Python GUI, and **upgraded design to 6-axis manipulator**.

Movie Recommendation System | Python, Pandas, NLTK, Scikit-learn

Jan 2024 – Apr 2024

- Built a content-based recommendation system by processing and vectorizing movie metadata, using cosine similarity to provide personalized recommendations.

Quadruped Robot | Python, Robotics, Inverse Kinematics

Jan 2024 – Apr 2024

- Engineered a quadruped robot by developing a Python-based inverse kinematics (IK) model to calculate joint angles for precise leg movement and control.

CERTIFICATIONS

- NPTEL - Robotics and Control
- Coding Ninjas - Java
- MATLAB - Image Processing
- Infosys - Web Development
- Altair - Machine Learning
- TensorFlow
- PyTorch
- OpenCV